EXPERIMENT-2

```
// Two-lane traffic light simulation for Proteus (Arduino Uno)
// Lane A: Red = pin 2, Yellow = pin 3, Green = pin 4
// Lane B: Red = pin 5, Yellow = pin 6, Green = pin 7

const int A_RED = 2;
const int A_YELLOW = 3;
const int A_GREEN = 4;

const int B_RED = 5;
const int B_YELLOW = 6;
const int B_GREEN = 7;

const unsigned long GREEN_TIME = 5000;  // ms
const unsigned long YELLOW_TIME = 2000; // ms

void setLane(int red, int yellow, int green, bool r, bool y, bool g) {
  digitalWrite(red, r ? HIGH : LOW);
  digitalWrite(yellow, y ? HIGH : LOW);
  digitalWrite(green, g ? HIGH : LOW);
}

void setup() {
  pinMode(A_RED, OUTPUT);
  pinMode(A_YELLOW, OUTPUT);
  pinMode(A_GREEN, OUTPUT);
  pinMode(B_RED, OUTPUT);
  pinMode(B_YELLOW, OUTPUT);
  pinMode(B_GREEN, OUTPUT);
```

```
  Serial.begin(9600);
  Serial.println("Traffic light system started");
}

void loop() {
  // State 1: Lane A green, Lane B red
  setLane(A_RED, A_YELLOW, A_GREEN, false, false, true);
  setLane(B_RED, B_YELLOW, B_GREEN, true, false, false);
  Serial.println("Lane A: GREEN | Lane B: RED");
  delay(GREEN_TIME);

  // State 2: Lane A yellow, Lane B red
  setLane(A_RED, A_YELLOW, A_GREEN, false, true, false);
  setLane(B_RED, B_YELLOW, B_GREEN, true, false, false);
  Serial.println("Lane A: YELLOW | Lane B: RED");
  delay(YELLOW_TIME);

  // State 3: Lane A red, Lane B green
  setLane(A_RED, A_YELLOW, A_GREEN, true, false, false);
  setLane(B_RED, B_YELLOW, B_GREEN, false, false, true);
  Serial.println("Lane A: RED | Lane B: GREEN");
  delay(GREEN_TIME);

  // State 4: Lane A red, Lane B yellow
  setLane(A_RED, A_YELLOW, A_GREEN, true, false, false);
  setLane(B_RED, B_YELLOW, B_GREEN, false, true, false);
  Serial.println("Lane A: RED | Lane B: YELLOW");
  delay(YELLOW_TIME);
}
```